

TEO LARA

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EDUCATION

University of Cambridge	Cambridge, UK
PhD Student in Applied Mathematics	May 2030
Massachusetts Institute of Technology	Cambridge, MA
Bachelor of Science in Mathematics & Physics	May 2026, GPA: 5.0/5.0

COURSEWORK

Maths: Numerical Methods for PDEs, Differential Geometry, Nonlinear Dynamics, Numerical Linear Algebra, Complex Analysis, Real Analysis, Differential Equations

Physics: Quantum Field Theory I & II, Quantum Mechanics I–III, Statistical Mechanics, General Relativity, Special Relativity, Experimental Physics, Classical Mechanics I–III

SKILLS

Software: Python, C++, Matlab, Mathematica, Bash, Git, Gnuplot, Paraview, VisIt

Languages: Fluent English, Fluent Spanish, Proficient French

RESEARCH EXPERIENCE

Dunkel Research Group, Cambridge, MA May 2025 - present
Advisors: Harry Walden, Ashutosh Tripathi, Jörn Dunkel

- Developing high-fidelity fluid–structure interaction simulations in IBAMR to analyze the biomechanics of jellyfish locomotion and feeding efficiency.

Kamrin Research Group, Cambridge, MA October 2023 - October 2026
Advisor: Ken Kamrin

- Creating advanced Eulerian computational frameworks to model soft solid–fluid contact interactions using reference map techniques.

Berkeley Lab Center for Computational Sciences, Berkeley, CA June - August 2024
Advisors: Saurabh Sawant, Andy Nonaka

- Implemented Non-Equilibrium-Green’s Function formalism to simulate quantum transport in carbon nanotubes.

Millholland Research Group, Cambridge, MA March - September 2023
Advisors: Sarah Millholland

- Investigated planetary system dynamics by modelling exoplanet resonance chains and obliquity evolution during migration phases using the REBOUNDx N-body integrator.

Bush Applied Maths Lab, Cambridge, MA November 2022 - September 2023
Advisors: Bauyrzhan Primkulov, John Bush

- Formulated a deep-learning image recognition pipeline to track and analyze trajectories of fluid droplets in “walking droplet” hydrodynamic quantum analog experiments.

PUBLICATIONS

1. **T. Lara**, K. Kamrin, “Fluid-immersed frictional contact of highly deformable solids in an Eulerian framework,” *Computer Methods in Applied Mechanics and Engineering*, vol. 463, 2027. doi: 10.1016/j.cma.2026.119415.

2. S. Sawant, **T. Lara**, et al., “Non-ideal subthreshold swing in aligned carbon nanotube transistors due to variable occupancy discrete charge traps,” *Physical Review Applied*, vol. 26, 2026. doi: 10.1103/m25k-qfgk.
3. **T. Lara**, K. Kamrin, “Unified Eulerian method for fluid-immersed self- and multi-body solid contact,” *Computer Methods in Applied Mechanics and Engineering*, vol. 437, 2025. doi: 10.1016/j.cma.2025.117745.
4. S. Millholland, **T. Lara**[†], J. Toomlaid[†], “Spin dynamics of planets in resonant chains,” *The Astrophysical Journal*, vol. 961, 2024. doi: 10.3847/1538-4357/ad10a0. († equal contribution)
5. **T. Lara**, A. Galvan, “Optimizing airfoil shape for small, low speed, unmanned gliders: A homemade investigation,” *Journal of Emerging Investigators*, vol. 6, 2023. doi: 10.59720/22-145.

PRESENTATIONS

- **T. Lara**, K. Kamrin, “Simulating Contact-Rich Fluid-Structure Interaction Problems,” US Association for Computational Mechanics Student Chapter Seminar Series, Webinar, August 2026.
- **T. Lara**, A. Tripathi, H. Walden, “The Fluid Dynamics of Feeding Jellyfish,” SPUR Student Project Seminar Series, Cambridge, MA, August 2025.
- **T. Lara**, K. Kamrin, “Unified Eulerian method for fluid-immersed self- and multi-body solid contact,” US National Congress on Computational Mechanics, Chicago, IL, July 2025.

HONOURS & AWARDS

NSF Graduate Research Fellowship Program Awardee	2026
Phi Beta Kappa Honour Society Member	2026
Sigma Pi Sigma Honour Society Member	2026
John Reed UROP Fund Recipient	2025
Paul E. Gray UROP Fund Recipient	2024
John P. Perry UROP Fund Recipient	2023
MIT SPUR Research Program Participant	2025
LBNL SULI Research Program Participant	2024

LEADERSHIP & INVOLVEMENT

MIT French House - President	2025 - 2026
MIT Strategic Games Society - Treasurer	2025 - 2026
MIT Climbing Team - Team Captain/Coach	2023 - 2025
MIT Video Game Orchestra - Arranging Chair	2022 - 2024
MIT Origami Society - Executive Member	2022 - 2023